Ahmedabad University

Subject - Digital Image Processing

8. Vehicle Orientation Estimation Using Conventional Image Processing

Weekly Report - 2

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Repo Link: https://github.com/encrypted-official/DIP_2026_Project8_Group1

Group - 1

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2026 (Monsoon Semester)

1. Work done this week

This week was mostly about getting to know the dataset properly and deciding how we are going to approach the problem. We did not start the actual implementation yet, as we felt it was better to be clear on a few basic points first.

We downloaded the DRASHTI-HaOBB dataset from Zenodo and went through its description along with the paper. We looked at a number of sample aerial images from the validation part to get an idea of what the vehicles look like from above. Most of them are quite small, some are partly covered by trees or shadows, and many are parked very close to each other. We also saw that every vehicle is marked with an oriented bounding box, a class name and a heading angle, so we will not need to label anything ourselves.

A good part of our discussion went into what exactly we should predict. The dataset heading tells the direction of the front of the vehicle, but most of the conventional methods we read about only find the main axis of the shape. They can say the car lies along a certain line, but not which end is the front. We have not fully fixed this yet, but we are leaning towards treating the axis as our main target and keeping full heading as an extra goal for later.

We also shortlisted the methods we want to try first. Based on what we read in Week 1 and this week, we picked PCA and the Hough transform as the first two, and kept image moments and minimum-area rectangle fitting as possible options for later. Finally, we divided the upcoming work among the four of us: one person on the dataset side, two on the two methods, and one on evaluation and results.

2. Work to be done next week

- Read the annotation files in detail and prepare vehicle crops from the given bounding boxes.
- Finalise how the heading angle will be converted into the axis angle we compare against.
- Start working on the first two methods, PCA and the Hough transform, on a small set of crops.
- Decide which measures we will use to compare the methods, such as average angle error and the share of vehicles that get a prediction.

3. Literature Review

This week we read a few more sources that relate to separating the vehicle from the background and fitting a shape to it, since most orientation methods depend on this step.

- **Gonzalez and Woods (2018)** cover the basic tools we expect to use, such as filtering, edge detection, thresholding and morphological operations. Going back to the textbook helped us see how these steps can be put together for small aerial objects.

- **Otsu (1979)** gave a simple way to pick a threshold automatically by looking at the grey level histogram. It could help in separating a vehicle from the road, though it may struggle when shadows or road paint are as dark or bright as the vehicle.
- **Suzuki and Abe (1985)** described the border following method that is used to trace the outline of objects in a binary image. Once a vehicle is separated, its outline can be used to study its shape and direction.
- **Freeman and Shapira (1975)** showed how to find the smallest rectangle that can enclose a closed shape at any angle. The longer side of such a rectangle should roughly follow the length of the vehicle, which is why we kept it as a possible method.

4. Challenges Faced

- The dataset is quite large, so downloading it and going through it took more time than we expected.
- Many vehicles are only a few pixels wide in the images, and it was hard to judge their direction even by eye in some cases.
- Shadows, trees, road markings and nearby vehicles often touch the vehicle, which will make it difficult to separate it from the background.
- We had some confusion on whether to predict the full heading or only the axis, and this is still being discussed within the group.

References

[1] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.

[2] N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," IEEE Transactions on Systems, Man, and Cybernetics, vol. 9, no. 1, pp. 62-66, 1979.

[3] S. Suzuki and K. Abe, "Topological Structural Analysis of Digitized Binary Images by Border Following," Computer Vision, Graphics, and Image Processing, vol. 30, no. 1, pp. 32-46, 1985.

[4] H. Freeman and R. Shapira, "Determining the Minimum-Area Encasing Rectangle for an Arbitrary Closed Curve," Communications of the ACM, vol. 18, no. 7, pp. 409-413, 1975.

[5] DRASHTI-HaOBB dataset, Zenodo, https://zenodo.org/records/18278989